Lecture15: Dimensionality Reduction 1

STA 4724

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Low-Dimensional Space in High-Dimensional Dataset

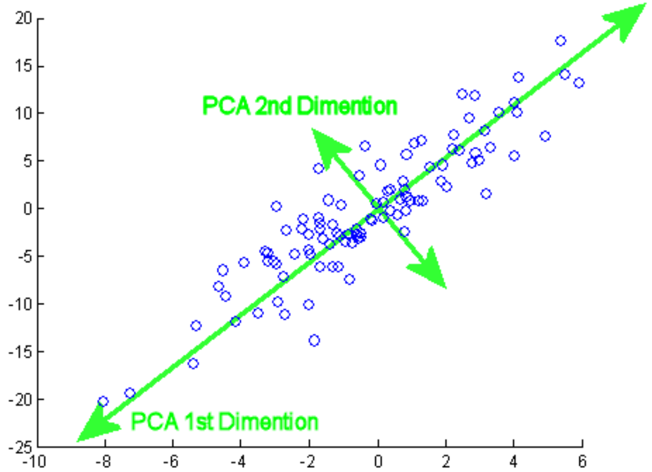
- One of the key tasks in big data analytics is pattern recognition. To the end, we implicitly assume there exist some patterns in dataset.
- If a large dataset can be described by a few patterns, then mathematically speaking, the data points are concentrated around some lower-dimensional space.
- Lasso regression that leads us to a sparse solution is one way to find a low-dimensional space, which is consist of the non-zero features.
- The process of finding low-dimensional space in high-dimensional dataset is called *dimensionality reduction*.
- In this session, we introduce another very important dimensionality reduction tool: Principal Component Analysis (PCA)

Principal Component Analysis

- Find the most significant component (weighted direction) that describes the dataset.

- Then, find the 2^{nd} most signification component that describes the rest of the dataset.

- Note that to describe the **rest** of the dataset, the 2^{nd} component is set perpendicular to the 1^{st} component.
  - In 2-dim world, there is only 1 direction perpendicular to the 1^{st} direction. However, in 3 or higher-dim world, there are infinitely many directions perpendicular to the 1^{st} direction.

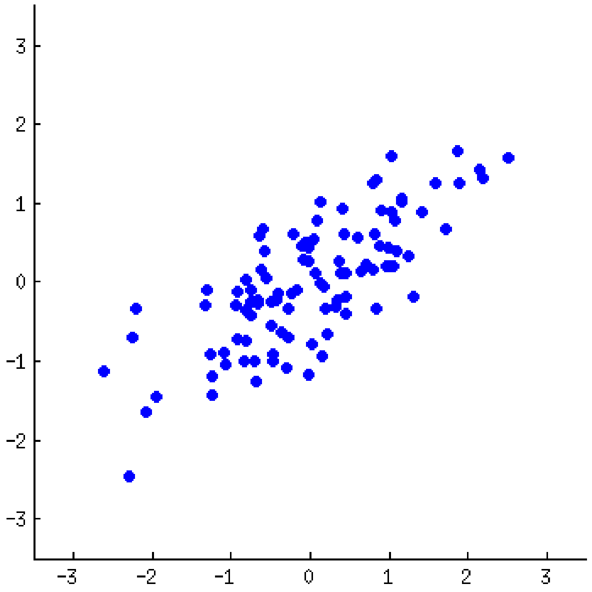


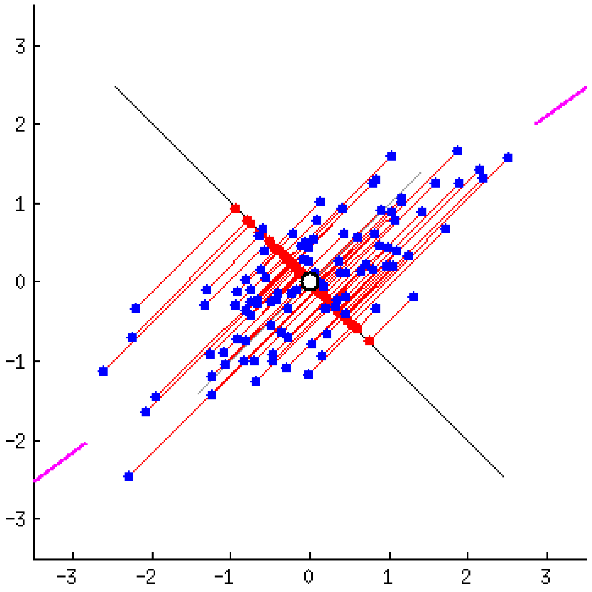
Principal Component Analysis

- By repeating the steps, we find n significant components for an n-feature dataset.
- We are essentially finding a new coordinate system that better represents the dataset.
- If we do a "sparse coding" that takes only the first few significant components, only the least important components are lost.
- That is, we have the dataset embedded in the low-dim subspace that is formed by the first few significant components. Therefore, dimensionality is reduced.

How to Find the Significant Components

- There are several different ways to find PCA.
- Least square is probably the least efficient yet easy-to-understand method.
- Use least square to find the most significant direction. Take that direction off from the dataset, and use least square to find 2^{nd} direction.
  - Remember a component is not only a direction.





How to Find the Significant Components

- Let's put the dataset in matrix form. An N–feature dataset consisting of M instances is written as a $M \times N$ matrix. We will call this matrix A.

- We can find the PCA via the eigenvalue decomposition of A's covariance matrix, AA^T.

- In this class, we don't have time to dig into the mathematical details behind the scene.

- At the high level, the covariance matrix provides the correlation among the data points. Eigenvalue decomposition extracts the pattern in the correlation and represent them as an orthogonal basis.

A Very Quick Review for Eigenvalue Decomposition

- For a symmetric $M \times M$ matrix B, eigenvalue decomposition of B is
$$B = Q\Lambda Q^{-1}$$
- Q is the square $n \times n$ matrix whose *i*-th column is the eigenvector q_i of B
- Λ is the diagonal matrix whose *i*-th diagonal element λ_i is the *i*-th eigenvalue

- For every *i*, we have
$$Bq_i = \lambda_i q_i$$

- Q is orthogonal matrix, i.e., $Q^T Q = I$

- This only apply to symmetric matrices, and covariance matrix is symmetric.

Time to Play

- We will introduce better approaches to finding PCA. For now, let's see how PCA works for dimensionality reduction.

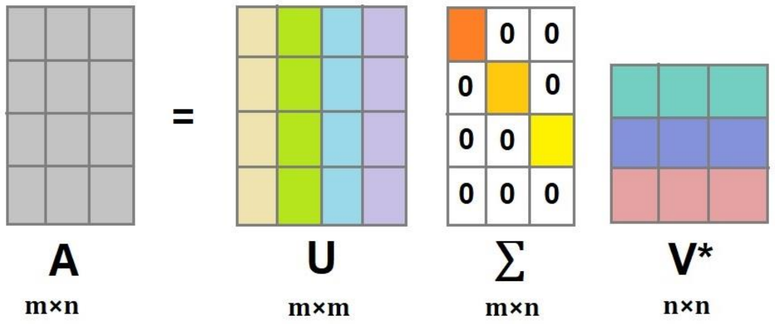
Singular Value Decomposition (SVD)

- Eigenvalue decomposition can be memory-intensive
  - The covariance matrix is unnecessarily huge when we have a lot of data points and a few features.
  - Covariance matrix of A is $M \times M$ even if $M \gg N$.

- Singular Value Decomposition (SVD) finds an equivalent decomposition without make the covariance matrix:

$$A = U\Sigma V^T$$

- U is $M \times M$ orthonormal matrix, i.e., $U^T U = I$
- V is $N \times N$ orthonormal matrix, i.e., $V^T V = I$
- Σ is $M \times N$ diagonal matrix whose elements are called the **singular values** of A



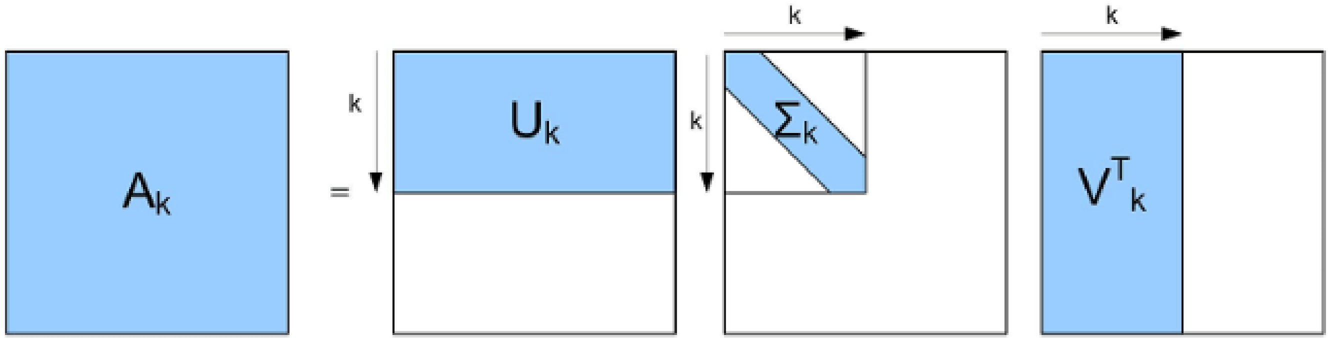
Dimensionality Reduction via SVD

- To reduce the dimensionality, we keep only first a few singular values and set the rest to 0.

- We can eliminate the columns corresponding to 0 singular values to save space. This is called compact SVD

- If keeps only the first k singular values, we call rank-k truncated SVD.

- Effectively, this is finding the first k significate components of A.



Time to Play

- Let’s see how to apply SVD in **python**!